



LoRA: Low-Rank Adaptation of Large Language Models : Rank Ablation Re-implementation Study

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Introduction

- **PROBLEM:** Fine-tuning large language models requires updating all parameters
- LoRA injects trainable low-rank matrices into frozen layers (no inference latency)

$$\mathbf{h} = \mathbf{W}_0 \mathbf{x} + \mathbf{B} \mathbf{A} \mathbf{x}$$

*A and B are initialized so $\Delta W = 0$ at the start of training.
At inference, they merge into W_0 with no overhead.*

Methodology

roberta-base (125M) used in place of GPT-3 (175B) due to compute constraints

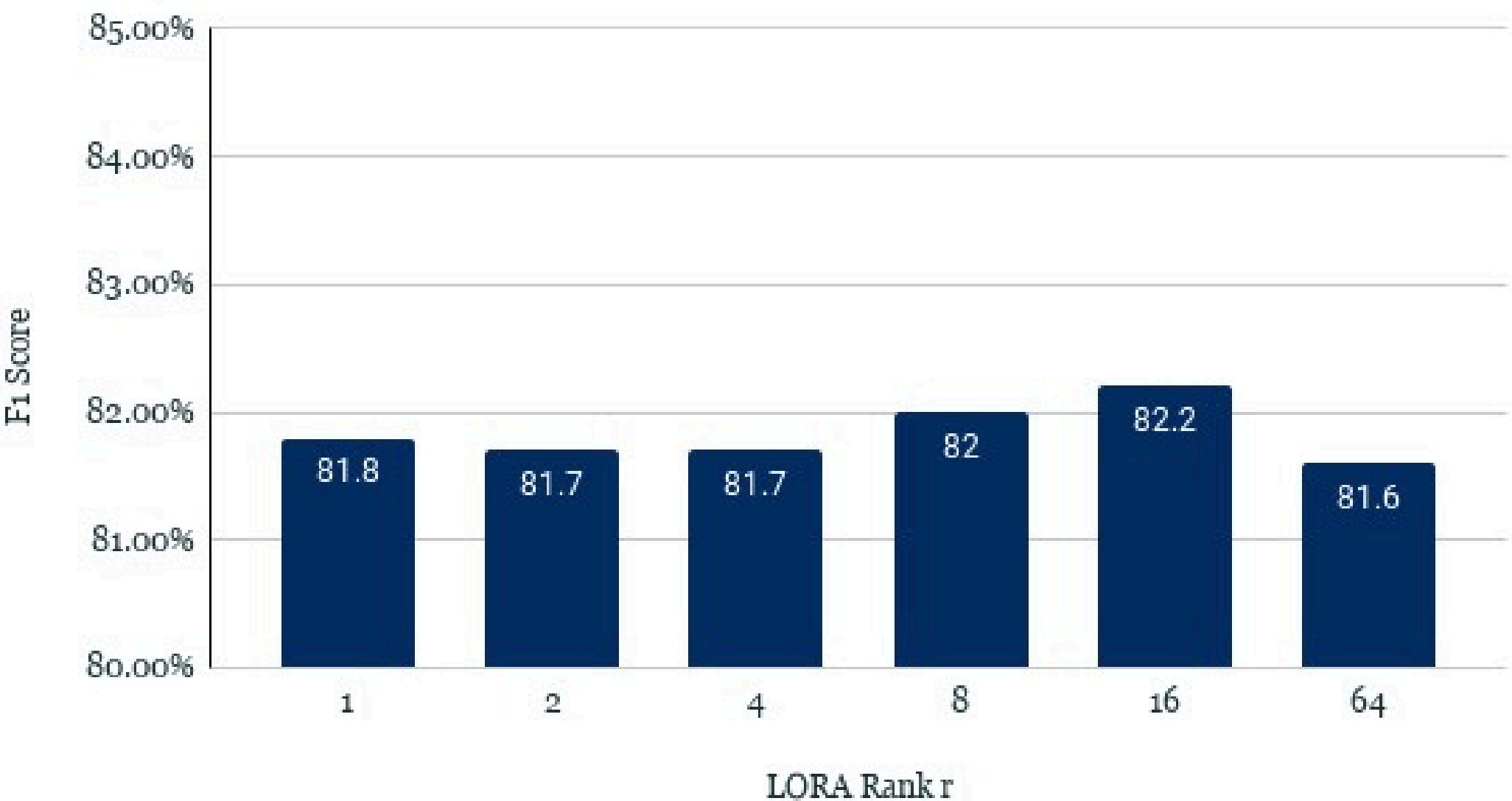
Parameter	Value
Base model	roberta-base (125M)
LoRA target	W_q, W_v
Dropout	0.1
Learning rate	3×10^{-4}
Batch size	32
Epochs	1
Ranks tested	1, 2, 4, 8, 16, 64

Conclusion

- Even with a much smaller model and limited compute, the rank ablation trend replicates at a fraction of the cost
- A single researcher can fine-tune a 125M parameter model in hours on CPU

Results

MRPC — F1 Score vs. Rank



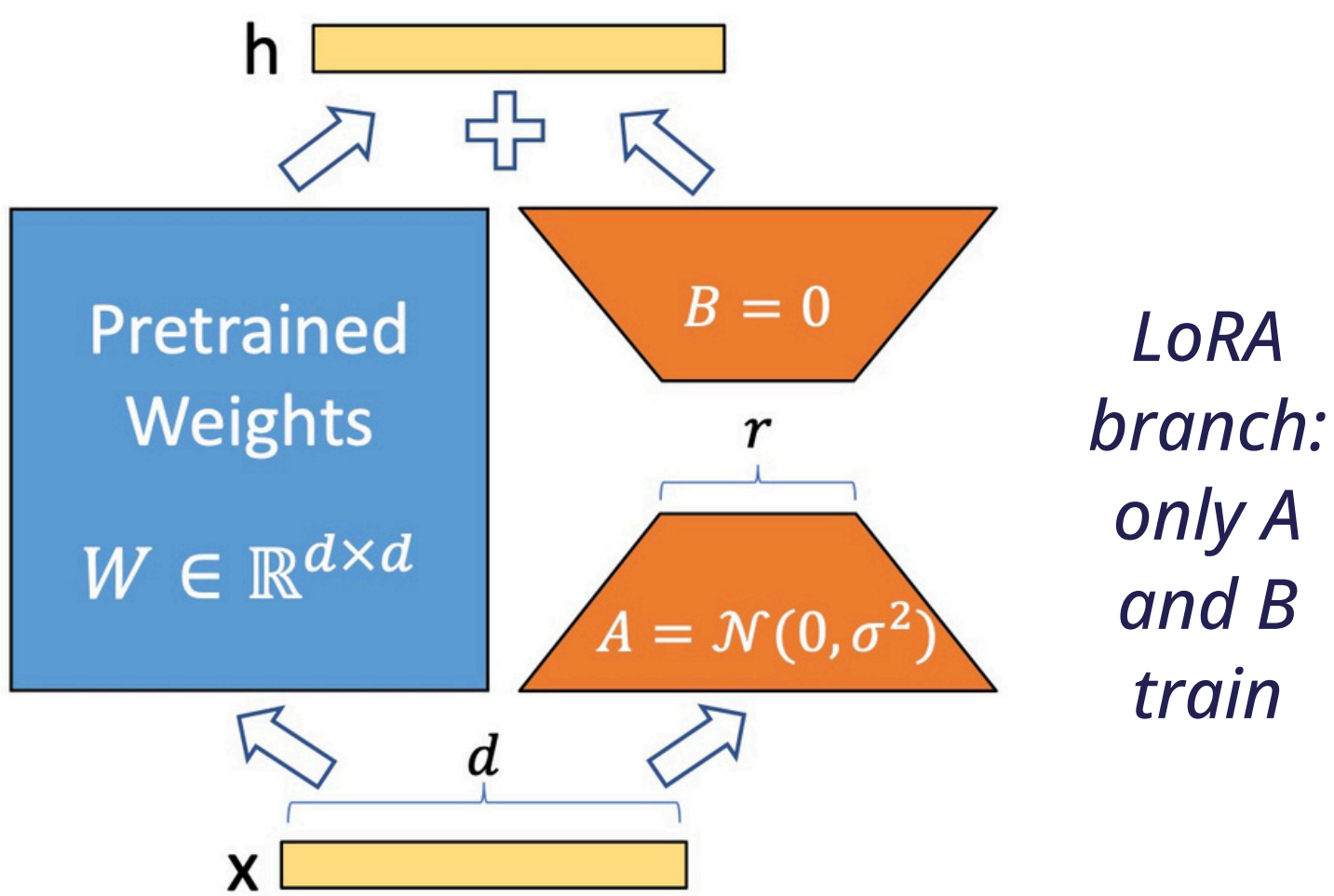
- SST-2 varies only 0.8% across all ranks
- $r=64$ uses 5× more params yet underperforms $r=1$ (92.8% vs. 93.6%)
- Our absolute scores are lower than the paper's (roberta-base vs. GPT-3, 1 epoch vs. multi-epoch), but the trend replicates cleanly.

SST2 — Accuracy vs. Rank



Observations

Rank r	SST-2 Acc.	MRPC F1	Params
$r = 1$	93.6% ★	81.80%	629K
$r = 2$	92.90%	81.70%	666K
$r = 4$	92.90%	81.70%	740K
$r = 8$	93.20%	82.00%	887K
$r = 16$	93.20%	82.2% ★	1.18M
$r = 64$	92.80%	81.60%	3.28M



Challenges

- Dependency conflicts required multiple debugging rounds before a single training run could complete.

Possible directions for future exploration:

- Train for more epochs to test if rank gaps widen with longer training
- Extend to GPT-2 or larger models to see if the same flat trend holds

References

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